US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12398598
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Nagasaka; Hidenori et al.

Earth auger

Abstract

A hole is formed in a drill target with improved workability. An electric work machine includes a motor, a motor housing accommodating the motor, a handle housing including a grip including a trigger switch operable to activate the motor, a reducer, a gear housing accommodating the reducer, a rotation output unit protruding frontward from the gear housing and rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit, and a handle fastened to the gear housing and at least partially located laterally from the gear housing.

Inventors: **Nagasaka; Hidenori (Anjo, JP), Imae; Tomohiko (Anjo, JP), Ito; Yoshihiro (Anjo, JP)**

Applicant: **MAKITA CORPORATION** (Anjo, JP)

Family ID: **1000008780598**

Assignee: **MAKITA CORPORATION (Anjo, JP)**

Appl. No.: **18/206692**

Filed: **June 07, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240044215 A1	Feb. 08, 2024

Foreign Application Priority Data

JP	2022-124099	Aug. 03, 2022

Publication Classification

Int. Cl.: **E21B11/00** (20060101); **E21B7/02** (20060101); **E21B7/04** (20060101)

U.S. Cl.:

CPC **E21B11/005** (20130101); **E21B7/027** (20130101); **E21B7/046** (20130101);

Field of Classification Search

CPC: E21B (11/005); B25F (5/026)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2020/0180124	12/2019	Ellice	N/A	B25F 5/026
2023/0082162	12/2022	Nagasaka	173/162.2	B23B 45/001

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202020102840	12/2019	DE	B25F 5/026
2020-196098	12/2019	JP	N/A

OTHER PUBLICATIONS

Translation for DE 202020102840 U1 (Year: 2025). cited by examiner

---

Primary Examiner: Wright; Giovanna

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2022-124099, filed on Aug. 3, 2022, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an earth auger.

2. Description of the Background

(3) In the technical field of earth augers, a known power tool drills the ground with a rotating earth auger drill bit, as described in Japanese Unexamined Patent Application Publication No. 2020-196098.

BRIEF SUMMARY

(4) An earth auger is used to form a lateral hole. For a piping operation to install sewage pipes underground, for example, a technique is awaited to form lateral holes with improved workability.

(5) The present disclosure is directed to forming a hole (in particular, a lateral hole) in a drill target with improved workability.

(6) A first aspect of the present disclosure provides an earth auger, including: a motor; a motor housing accommodating the motor; a handle housing including a grip including a trigger switch

operable to activate the motor; a reducer; a gear housing accommodating the reducer; a rotation output unit protruding frontward from the gear housing, the rotation output unit being rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit; and a handle fastened to the gear housing and at least partially located laterally from the gear housing.
(7) A second aspect of the present disclosure provides an earth auger, including: a motor; a motor housing accommodating the motor; a handle housing including a grip including a trigger switch operable to activate the motor; a reducer; a gear housing accommodating the reducer; a rotation output unit protruding frontward from the gear housing, the rotation output unit being rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit; and a contact member located below the rotation output unit and at least partially located laterally from the gear housing.
(8) The technique according to the above aspects of the present disclosure is used to form a hole (in particular, a lateral hole) in a drill target with improved workability.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a front perspective view of an earth auger according to a first embodiment.
(2) FIG. **2** is an exploded front perspective view of the earth auger according to the first embodiment.
(3) FIG. **3** is a side view of a body in the earth auger according to the first embodiment.
(4) FIG. **4** is a sectional view of the body in the earth auger according to the first embodiment.
(5) FIG. **5** is a side view of the earth auger according to the first embodiment, describing its use.
(6) FIG. **6** is a perspective view of the earth auger according to the first embodiment, describing its use.
(7) FIG. **7** is a perspective view of an earth auger according to a second embodiment, describing its use.
(8) FIG. **8** is a perspective view of an earth auger according to a third embodiment, describing its use.
(9) FIG. **9** is a front view of an earth auger according to a fourth embodiment, describing its use.
(10) FIG. **10** is a perspective view of the earth auger according to the fourth embodiment, describing its use.
(11) FIG. **11** is a perspective view of an earth auger according to a fifth embodiment, describing its use.
(12) FIG. **12** is a perspective view of an additional handle in the fifth embodiment.
(13) FIG. **13** is a front perspective view of an earth auger according to a sixth embodiment.
DETAILED DESCRIPTION
(14) Although one or more embodiments of the present disclosure will now be described with reference to the drawings, the present disclosure is not limited to the present embodiments. The components in the embodiments described below may be combined as appropriate. One or more components may be eliminated.
(15) In the embodiments, the positional relationships between the components will be described using the directional terms such as right and left (or lateral), front and rear (or frontward and rearward), and up and down (or vertical). The terms indicate directions of an earth auger **1** during its use. For example, the earth auger **1** for sale at a store or in a sales catalog may be displayed upside down.
First Embodiment
Electric Work Machine

(16) FIG. **1** is a front perspective view of the earth auger **1** according to the present embodiment. FIG. **2** is an exploded front perspective view of the earth auger **1**.
(17) The earth auger **1** is a portable earth auger being held by an operator for drilling a drill target to form a hole in the drill target. The drill target is the ground in the present embodiment. The earth auger **1** is used to form a lateral hole in the ground. The earth auger **1** includes a body **6**, a drill bit **19**, an additional handle **20**, and a contact member **22**.
(18) FIG. **3** is a side view of the body **6** of the earth auger **1**. FIG. **4** is a sectional view of the body **6** in the earth auger **1**.
(19) As shown in FIGS. **1** to **4**, the body **6** includes a motor housing **2**, a handle housing **3**, a gear housing **4**, battery mounts **7**, a controller **13**, a main switch **10**, a trigger switch **11**, a forward-reverse switch lever **9**, a speed switch lever **15**, a motor **8**, a reducer **14**, and a rotation output unit **5**.
(20) The motor housing **2** accommodates the motor **8**. The motor housing **2** is cylindrical and extends vertically. The motor housing **2** is formed from a synthetic resin. The motor housing **2** has outlets **2**E. The outlets **2**E connect the inside and the outside of the motor housing **2**. The motor housing **2** has the outlets **2**E in its left, right, and front. Air in an internal space of the motor housing **2** is discharged out of the motor housing **2** through the outlets **2**E.
(21) The handle housing **3** is located above the motor housing **2**. The handle housing **3** includes a lower portion connected to an upper portion of the motor housing **2**. The handle housing **3** is a vertically elongated loop. The handle housing **3** is formed from a synthetic resin. The handle housing **3** includes a front portion **3**A, a grip **3**B, a controller compartment **3**C, and a battery connector **3**D. The front portion **3**A is connected to the upper portion of the motor housing **2**. The grip **3**B extends upward from the rear of the front portion **3**A. The controller compartment **3**C extends upward from the front of the front portion **3**A. The battery connector **3**D connects the upper end of the grip **3**B and the upper end of the controller compartment **3**C. The grip **3**B is located rearward from the controller compartment **3**C. The grip **3**B is located above the motor housing **2**. The operator may grip the grip **3**B with a hand. The controller compartment **3**C has an inlet **3**F at the front.
(22) The gear housing **4** accommodates the reducer **14**. The gear housing **4** is cylindrical and extends vertically. The gear housing **4** is located below the motor housing **2**. The gear housing **4** includes its upper portion connected to a lower portion of the motor housing **2**. The gear housing **4** is formed from aluminum. The gear housing **4** has its surface at least partially covered with a cover **4**A. The cover **4**A in the embodiment has a two-layer structure of a synthetic resin and an elastomer.
(23) The gear housing **4** has threaded holes **16** in its right and left portions. The threaded hole **16** are vertically aligned with the rotation output unit **5**.
(24) The gear housing **4** has threaded holes **18** in its right and left portions. The threaded holes **18** are located behind and below the corresponding threaded holes **16**.
(25) The battery mounts **7** are located in an upper portion of the handle housing **3**. The battery mounts **7** receive battery packs **17**. The battery mounts **7** are located on the battery connector **3**D in the handle housing **3**. The body **6** in the present embodiment includes two battery mounts **7** arranged in the front-rear direction. The two battery mounts **7** each receive the corresponding battery pack **17**. The two battery packs **17** are thus arranged in the front-rear direction. The battery packs **17** are detachable from the battery mounts **7**. The battery packs **17** attached to the battery mounts **7** power the earth auger **1**. Each battery pack **17** includes a secondary battery. The battery pack **17** in the present embodiment includes a rechargeable lithium-ion battery.
(26) The controller **13** outputs a control signal for controlling the earth auger **1**. The controller compartment **3**C has an internal space that can accommodate the controller **13**. The controller **13** is accommodated in the controller compartment **3**C.
(27) The main switch **10** is operable by the operator to activate the earth auger **1**. The main switch

10 is located on the rear of the front portion **3**A. The main switch **10** causes the battery packs **17** to supply power to the controller **13**. This activates the earth auger **1**. The main switch **10** is operable to switch the earth auger **1** between the activated state and the stopped state.

(28) The trigger switch **11** is operable by the operator to activate the motor **8**. The trigger switch **11** is located in the grip **3**B. The trigger switch **11** includes a trigger lever **11**A and a switch circuit **11**B. The trigger lever **11**A protrudes frontward from a lower front portion of the grip **3**B. The operator holding the grip **3**B with the right or left hand operates the trigger lever **11**A with a finger to move the trigger lever **11**A backward. The grip **3**B has an internal space that can accommodate the switch circuit **11**B. The switch circuit **11**B is accommodated in the grip **3**B. In response to an operation on the trigger lever **11**A, the switch circuit **11**B outputs an operation signal. When the trigger lever **11**A is pulled backward with the earth auger **1** being activated, the battery packs **17** supply power to the motor **8** and activate the motor **8**. The motor **8** is driven in response to the operation signal output from the switch circuit **11**B. The trigger lever **11**A is operable to switch between the operation state and the release state to switch the motor **8** between the driving state and the stopped state.

(29) The forward-reverse switch lever **9** is operable by the operator to switch the rotation direction of the motor **8**. The forward-reverse switch lever **9** is located in the front portion **3**A. The forward-reverse switch lever **9** is operable rightward or leftward to switch the rotation direction of the motor **8** between forward and reverse. This can switch the rotation direction of the rotation output unit **5** between forward and reverse.

(30) The speed switch lever **15** is operable by the operator to switch the rotational speed of the rotation output unit **5**. The speed switch lever **15** is located at the front of the gear housing **4**. An operation on the speed switch lever **15** in the vertical direction switches the rotational speed of the rotation output unit **5** between a first speed and a second speed. The second speed is higher than the first speed.

(31) The motor **8** generates a rotational force for rotating the rotation output unit **5**. The motor **8** is driven by power supplied from the battery packs **17**. The motor **8** is an inner-rotor brushless motor. The motor **8** includes a cylindrical stator **81** and a rotor **82** located inward from the stator **81**. The rotor **82** has a rotation axis extending vertically.

(32) The stator **81** includes a stator core **81**A, a first insulator **81**B, a second insulator **81**C, multiple coils **81**D, a sensor circuit board **81**E, and a connection wire **81**F. The stator core **81**A includes multiple steel plates stacked on one another. The first insulator **81**B is located in a lower portion of the stator core **81**A. The second insulator **81**C is located in an upper portion of the stator core **81**A. The coils **81**D are wound around the stator core **81**A with the first insulator **81**B and the second insulator **81**C in between. The sensor circuit board **81**E is attached to the second insulator **81**C. The connection wire **81**F is supported by the second insulator **81**C. The sensor circuit board **81**E includes multiple rotation detectors to detect rotation of the rotor **82**.

(33) The rotor **82** includes a rotor shaft **82**A, a rotor core **82**B, and multiple permanent magnets **82**C. The rotor core **82**B is cylindrical and surrounds the rotor shaft **82**A. The permanent magnets **82**C are held by the rotor core **82**B. The rotor shaft **82**A includes a lower portion rotatably supported by a bearing **83**. The rotor shaft **82**A includes an upper portion rotatably supported by a bearing **84**.

(34) A centrifugal fan **85** is mounted on a part of the rotor shaft **82**A between the bearing **83** and the stator **81**. The outlets **2**E in the motor housing **2** are located to partially surround the centrifugal fan **85**. As the rotor shaft **82**A rotates and thus the centrifugal fan **85** rotates, air in the internal space of the motor housing **2** is discharged out of the motor housing **2** through the outlets **2**E.

(35) The rotor shaft **82**A receives a pinion gear **141**S on its lower end. The pinion gear **141**S is located in an internal space of the gear housing **4**. The rotor shaft **82**A is connected to the reducer **14** with the pinion gear **141**S in between.

(36) The reducer **14** transmits a rotational force generated by the motor **8** to the rotation output unit

5. The reducer **14** transmits the rotational force through the rotor shaft **82**A to the rotation output unit **5**. The reducer **14** includes multiple gears. The reducer **14** includes a first planetary gear assembly **141**, a second planetary gear assembly **142**, a countershaft **143**, and an output shaft **144**.
(37) The first planetary gear assembly **141** is located below the rotor shaft **82**A. The countershaft **143** is located below the first planetary gear assembly **141**. The second planetary gear assembly **142** is located below the countershaft **143**. The output shaft **144** is located below the second planetary gear assembly **142**.
(38) The first planetary gear assembly **141** includes the pinion gear **141**S, multiple planetary gears **141**P, a first carrier **141**C, an internal gear **141**R, and a support pin **145**. The pinion gear **141**S serves as a sun gear. The planetary gears **141**P surround the pinion gear **141**S. The first carrier **141**C supports the planetary gears **141**P in a rotatable manner. The internal gear **141**R surrounds the planetary gears **141**P. The support pin **145** is held by the first carrier **141**C.
(39) The pinion gear **141**S is located at the lower end of the rotor shaft **82**A. The planetary gears **141**P mesh with the pinion gear **141**S and the internal gear **141**R. The first carrier **141**C holds the support pin **145**. The support pin **145** extends vertically. The support pin **145** is connected to the planetary gears **141**P. The first carrier **141**C supports the planetary gears **141**P in a rotatable manner with the support pin **145**.
(40) The second planetary gear assembly **142** includes a sun gear **142**S, multiple planetary gears **142**P, a second carrier **142**C, an internal gear **142**R, and a support pin **146**. The planetary gears **142**P surround the sun gear **142**S. The second carrier **142**C supports the planetary gears **142**P in a rotatable manner. The internal gear **142**R surrounds the planetary gears **142**P. The support pin **146** is held by the second carrier **142**C.
(41) The sun gear **142**S is located at the lower end of the countershaft **143**. The planetary gears **142**P mesh with the sun gear **142**S and the internal gear **142**R. The second carrier **142**C holds the support pin **146**. The support pin **146** extends vertically. The support pin **146** protrudes upward from the second carrier **142**C. The support pin **146** supports the planetary gears **142**P in a rotatable manner. The support pin **146** has its upper end protruding upward from the planetary gears **142**P. The second carrier **142**C supports the planetary gears **142**P in a rotatable manner with the support pin **146**.
(42) The internal gear **141**R in the first planetary gear assembly **141** is fixed to the gear housing **4**. The internal gear **141**R does not rotate. The internal gear **142**R in the second planetary gear assembly **142** is rotatable.
(43) The output shaft **144** is rotatably supported by the bearing **147**. The output shaft **144** has its upper end fixed to the second carrier **142**C. The output shaft **144** receives a bevel gear **148** at its lower end. The output shaft **144** has the lower end connected to the rotation output unit **5** with the bevel gear **148** in between. As the second carrier **142**C rotates, the second carrier **142**C and the output shaft **144** rotate together.
(44) The rotation axis of the rotor shaft **82**A, the rotation axis of the first carrier **141**C, the rotation axis of the countershaft **143**, the rotation axis of the second carrier **142**C, and the rotation axis of the output shaft **144** are aligned with one another.
(45) The reducer **14** includes a switching member **150**. The switching member **150** is movable vertically between the first planetary gear assembly **141** and the second planetary gear assembly **142**. The switching member **150** surrounds the countershaft **143**. The switching member **150** is connected to the speed switch lever **15**. In response to an operation on the speed switch lever **15**, the switching member **150** moves vertically.
(46) The switching member **150** is located below the first carrier **141**C. The support pin **145** includes a lower portion protruding downward from the first carrier **141**C. The switching member **150** has a hole **150**H. The support pin **145** protruding downward from the first carrier **141**C is placed in the hole **150**H. The switching member **150** is movable vertically while being guided by the support pin **145**. This switches the rotational speed of the rotation output unit **5**.

(47) The reducer **14** includes a connector **151**. The connector **151** is located in an upper portion of the planetary gear **142**P. The connector **151** has a hole **151**H. The support pin **146** protruding upward from the planetary gears **142**P is placed in the hole **151**H. The connector **151** is connected to the second carrier **142**C with the support pin **146**.
(48) The switching member **150** is movable between a first position and a second position. The second position is below the first position. The switching member **150** is movable between the first position and the second position while being guided by the support pin **145**.
(49) At the first position, the switching member **150** is connected to the first carrier **141**C and the countershaft **143**. At the first position, the switching member **150** is apart from the connector **151**. At the first position, the switching member **150** is integral with the first carrier **141**C and the upper end the countershaft **143**. At the first position, the switching member **150** rotates together with the first carrier **141**C and the countershaft **143** as the first carrier **141**C rotates.
(50) At the second position, the switching member **150** is connected to the connector **151**. At the second position, the switching member **150** is apart from the first carrier **141**C and the countershaft **143**. With the switching member **150** at the second position, the upper end of the countershaft **143** and the first carrier **141**C are apart. At the second position, the switching member **150** is integral with the connector **151**. The support pin **145** is placed in the hole **150**H in the switching member **150**. The support pin **145** is connected to the planetary gears **141**P. At the second position, the switching member **150** rotates together with the connector **151** as the planetary gear **141**P revolves.
(51) At the first position, the switching member **150** is connected to the countershaft **143** with splines. At the second position, the switching member **150** is disconnected from the countershaft **143** in response to disengagement of the splines.
(52) With the switching member **150** at the first position, the pinion gear **141**S rotates and the planetary gears **141**P revolve about the pinion gear **141**S as the motor **8** drives the rotor shaft **82**A to rotate. At the first position, the switching member **150** is integral with the first carrier **141**C and the upper end of the countershaft **143**. As the planetary gears **141**P revolve, the first carrier **141**C, the countershaft **143**, and the switching member **150** rotate together at a lower rotational speed than the rotor shaft **82**A. As the countershaft **143** rotates, the sun gear **142**S rotates. The planetary gears **142**P then revolve about the sun gear **142**S. The second carrier **142**C and the output shaft **144** then rotate at a lower rotational speed than the countershaft **143**. With the switching member **150** at the first position, both the first planetary gear assembly **141** and the second planetary gear assembly **142** operate for rotation reduction as the motor **8** drives, thus causing the output shaft **144** to rotate at the first speed.
(53) With the switching member **150** at the second position, the pinion gear **141**S rotates and the planetary gears **141**P revolve about the pinion gear **141**S as the motor **8** drives the rotor shaft **82**A to rotate. The first carrier **141**C then rotates at a lower rotational speed than the rotor shaft **82**A. At the second position, the switching member **150** is apart from the first carrier **141**C and the countershaft **143**. With the switching member **150** at the second position, the upper end of the countershaft **143** and the first carrier **141**C are apart. At the second position, the switching member **150** is integral with the connector **151**. With the switching member **150** at the second position, the support pin **145** is placed in the hole **150**H in the switching member **150**. The revolving planetary gears **141**P cause the connector **151** and the switching member **150** to rotate at the same rotational speed as the first carrier **141**C. The rotating connector **151** causes the planetary gears **142**P to revolve at a revolution speed that is the same as the rotational speed of the connector **151**. The revolving planetary gears **142**P cause the second carrier **142**C and the output shaft **144** to rotate at the same rotational speed. With the switching member **150** at the second position, the first planetary gear assembly **141** operates for rotation reduction as the motor **8** drives, but the second planetary gear assembly **142** does not operate for rotation reduction. The output shaft **144** thus rotates at the second speed.
(54) The rotation output unit **5** rotates under a rotational force transmitted from the motor **8** through

the reducer **14**. The rotation output unit **5** protrudes frontward from the lower portion of the gear housing **4**. The rotation output unit **5** receives a drill bit. The rotation output unit **5** with the drill bit being attached is rotatable. The rotation output unit **5** has a rotation axis extending in the front-rear direction. The motor **8** has the rotation axis orthogonal to the rotation axis of the rotation output unit **5**.
(55) The rotation output unit **5** includes a spindle **51** and a drill chuck **52**. The drill chuck **52** is attached to the front end of the spindle **51**. The drill chuck **52** has an insertion hole **52**A. The drill bit is placed in the insertion hole **52**A. The insertion hole **52**A extends rearward from the front end of the drill chuck **52**. The drill chuck **52** with the drill bit **19** being attached is rotatable.
(56) The spindle **51** is rotatably supported by a needle bearing **53** and a ball bearing **54**. The needle bearing **53** supports the rear end of the spindle **51** in a rotatable manner. The ball bearing **54** supports the front of the spindle **51** in a rotatable manner.
(57) The spindle **51** receives a bevel gear **55** at its rear. The bevel gear **55** meshes with the bevel gear **148** on the output shaft **144**. The bevel gear **55** has a larger diameter than the bevel gear **148**. The bevel gear **55** includes more teeth than the bevel gear **148**.
(58) The rotation output unit **5** receives the drill bit **19**. The drill bit **19** is an earth auger drill bit used for drilling. The drill bit **19** includes a drill shaft **19**A, a double-helix drill blade **19**B, a tip bit **19**C, and two cutting blades **19**D.
(59) The drill shaft **19**A extends in the front-rear direction. To attach the drill bit **19** to the rotation output unit **5**, an adapter **5**A is placed in the insertion hole **52**A in the drill chuck **52**. The adapter **5**A is a rod-like member. The drill shaft **19**A includes a hole to receive the adapter **5**A at its rear end. With the adapter **5**A placed in the hole at the rear end of the drill shaft **19**A, the rear end of the drill shaft **19**A and the adapter **5**A are fastened with a fastener **5**B. The rear end of the drill shaft **19**A is attached to the drill chuck **52** using the adapter **5**A.
(60) The double-helix drill blade **19**B spirally surrounds the drill shaft **19**A. The double-helix drill blade **19**B is fixed to the drill shaft **19**A. The tip bit **19**C is at the front end of the drill shaft **19**A. Each of the two cutting blades **19**D is located at the front end of the double-helix drill blade **19**B.
(61) The additional handle **20** is fastened to the gear housing **4** with handle joints **21**. The handle joints **21** are located on the left and right of the gear housing **4**. Each handle joint **21** includes a base **211** and a fixing portion **212**. The base **211** includes a screw opening **21**C and a screw opening **21**D. The fixing portion **212** has screw openings **21**B. A screw placed in the screw opening **21**C is screwed into the threaded hole **16**. A screw placed in the screw opening **21**D is screwed into the threaded hole **18**. The base **211** in the handle joint **21** is thus fastened to the gear housing **4**.
(62) The additional handle **20** is at least partially located laterally from the gear housing **4**. Being laterally from the gear housing **4** refers to being either leftward or rightward or both from the gear housing **4**. In the present embodiment, the additional handle **20** is partially located leftward from the gear housing **4**. The additional handle **20** is partially located rightward from the gear housing **4**.
(63) The additional handle **20** is formed from a hollow metal pipe that is bent. The additional handle **20** includes insertion portions **20**C, vertical rods **20**B, and a lateral rod **20**A. Each insertion portion **20**C is fixed between the base **211** and the fixing portion **212** of the handle joint **21**. Each vertical rod **20**B extends upward from the insertion portion **20**C. The lateral rod **20**A extends laterally. The additional handle **20** includes a pair of insertion portions **20**C. The insertion portions **20**C in the pair are at an interval from each other in the lateral direction. The additional handle **20** includes a pair of vertical rods **20**B. The vertical rods in the pair are at an interval from each other in the lateral direction. Each vertical rod **20**B includes an upper portion sloping upward toward the front. The lateral rod **20**A connects the upper ends of the vertical rods **20**B in the pair.
(64) The insertion portion **20**C has multiple screw openings. The screw openings in the insertion portion **20**C are at an interval from each other in the vertical direction. The screw openings **21**B are located in the fixing portion **212** of the handle joint **21**. Threaded holes **21**A are located in the base **211** in the handle joint **21**. With the insertion portion **20**C between the base **211** and the fixing

portion **212**, a screw placed in the screw opening **21**B is screwed into the threaded hole **21**A through the screw opening in the insertion portion **20**C. The additional handle **20** is thus fastened to the handle joints **21**.
(65) The additional handle **20** is fastened to the handle joints **21**, and the handle joints **21** are fastened to the gear housing **4**. The additional handle **20** is thus fastened to the gear housing **4** with the handle joints **21** in between. In this state, the vertical rods **20**B extend upward from lateral portions of the gear housing **4**. One vertical rod **20**B extends upward from the left portion of the gear housing **4**. The other vertical rod **20**B extends upward from the right portion of the gear housing **4**. The lateral rod **20**A is located frontward from the gear housing **4**.
(66) The contact member **22** is located below the rotation output unit **5**. The contact member **22** is at least partially located laterally from the gear housing **4**. The contact member **22** has its left portion located leftward from the gear housing **4**. The contact member **22** has its right portion located rightward from the gear housing **4**. In other words, the contact member **22** has its left end located leftward from the left surface of the gear housing **4**. The contact member **22** has its right end located rightward from the right surface of the gear housing **4**.
(67) The contact member **22** is fastened to the gear housing **4** with the handle joints **21** in between. Each base **211** in the corresponding handle joint **21** has an insertion hole (not shown) extending upward from the lower surface of the base **211**. The contact member **22** includes a pair of insertion portions **22**C. The insertion portions **22**C are placed in the insertion holes in the base **211**. The insertion portions **22**C in the pair are arranged at an interval from each other in the lateral direction. Each insertion portion **22**C has multiple threaded holes **22**D. The threaded holes **22**D are at an interval from each other in the vertical direction. Each handle joint **21** has screw openings **21**E. With the insertion portions **22**C placed in second insertion holes, a screw placed in the corresponding screw opening **21**E is screwed into the corresponding threaded hole **22**D. The contact member **22** is thus fastened to the handle joints **21**.
(68) The contact member **22** is fastened to the handle joints **21**, and the handle joints **21** are fastened to the gear housing **4**. The contact member **22** is thus fastened to the gear housing **4** with the handle joints **21** in between. In this state, a contact surface **22**A of the contact member **22** faces downward, and a front surface **22**B of the contact member **22** faces frontward.
Method of Use
(69) A method of use of the earth auger **1** according to the present embodiment will now be described. FIG. **5** is a side view of the earth auger **1** according to the present embodiment, describing its use. FIG. **6** is a perspective view of the earth auger **1**, describing its use. The earth auger **1** is used to form a lateral hole in a drill target. The earth auger **1** is used with the motor housing **2** located above the gear housing **4**. In other words, the earth auger **1** is used in a vertical position. With the motor housing **2** located above the gear housing **4** in the vertical position, the contact member **22** is fixed to the gear housing **4**. The contact member **22** is in contact with the ground.
(70) The operator grips the additional handle **20** with the left hand and the grip **3**B in the handle housing **3** with the right hand. The operator grips the left of the lateral rod **20**A or the left vertical rod **20**B with the left hand. The operator operates the trigger lever **11**A with, for example, a little finger of the right hand holding the grip **3**B. This causes the rotation output unit **5** to rotate in a rotation direction indicated by the arrow in FIG. **6**, with the drill bit **19** being attached.
(71) With the contact surface **22**A in contact with the ground, the operator presses a front end **19**U of the rotating drill bit **19** against the drill target in front of the earth auger **1**. This allows drilling into the drill target to form a lateral hole in the drill target. The operator pushes the earth auger **1** forward with the drill bit **19** rotating. The contact member **22** may slide on the ground. This allows the earth auger **1** to move forward smoothly. To discharge a drilled material from the lateral hole formed in the drill target, the operator pulls out the earth auger **1** backward. The earth auger **1** moves backward smoothly as well.

(72) The operator may grip the additional handle **20** with the right hand and the grip **3**B in the handle housing **3** with the left hand. The operator may grip the right of the lateral rod **20**A or the right vertical rod **20**B with the right hand.
(73) During a drilling operation into the drill target, a large reaction force may act on the gear housing **4** through the drill bit **19** and the rotation output unit **5**. When the drill bit **19** rotates in the rotation direction indicated by the arrow in FIG. **6**, a reaction force may act to cause the upper end of the body **6** to fall leftward.
(74) In the present embodiment, the left portion of the contact member **22** is located leftward from the gear housing **4**. The left portion of the contact member **22** is in contact with the ground, receiving a reaction force acting on the gear housing **4**.
(75) The additional handle **20** gripped by the operator receives a reaction force acting on the gear housing **4**.
(76) The grip **3**B is gripped by the operator. The handle housing **3** thus receives a reaction force acting on the gear housing **4**.
(77) As described above, the earth auger **1** includes the motor **8**, the motor housing **2** accommodating the motor **8**, the handle housing **3** including the grip **3**B including the trigger switch **11** operable to activate the motor **8**, the reducer **14**, the gear housing **4** accommodating the reducer **14**, the rotation output unit **5** protruding frontward from the gear housing **4** and rotatable under a rotational force transmitted from the motor **8** through the reducer **14** with the drill bit **19** being attached to the rotation output unit **5**, and the additional handle **20** fastened to the gear housing **4** and at least partially located laterally from the gear housing **4**.
(78) With the above structure, the operator can perform a drilling operation while gripping the additional handle **20** with one hand and the grip **3**B in the handle housing **3** with the other hand. This allows formation of a lateral hole in the drill target with improved workability.
Second Embodiment
(79) A second embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.
(80) FIG. **7** is a perspective view of an earth auger **1** according to the present embodiment, describing its use. In the present embodiment, the contact member **22** and the handle joints **21** are eliminated.
(81) A loop-shaped additional handle **23** is fastened to the left portion of the gear housing **4**. The operator can grip any part of the additional handle **23**. The additional handle **23** includes a facing portion **23**B, a grip portion **23**E, an upper joint **23**C, and a lower joint **23**D. The facing portion **23**B faces the left surface of the gear housing **4**. The grip portion **23**E is located leftward from the facing portion **23**B. The upper joint **23**C connects an upper portion of the facing portion **23**B to an upper portion of the grip portion **23**E. The lower joint **23**D connects a lower portion of the facing portion **23**B to a lower portion of the grip portion **23**E. The facing portion **23**B has a screw opening **23**A. A screw placed in the screw opening **23**A is screwed into the threaded hole **18**. The additional handle **23** is thus fastened to the gear housing **4**.
(82) As in the first embodiment described above, the earth auger **1** is used in a vertical position with the motor housing **2** located above the gear housing **4**. In this state, the lower end of the gear housing **4** is flush with at least a part of the additional handle **23**. In the embodiment, the lower end of the gear housing **4** is flush with the lower end of the lower joint **23**D. With the lower end of the gear housing **4** in contact with the ground, the additional handle **23** is at least partially in contact with the ground. In the embodiment, the lower end of the lower joint **23**D is in contact with the ground.
(83) The operator grips the grip portion **23**E in the additional handle **23** with the left hand and the grip **3**B in the handle housing **3** with the right hand. The operator operates the trigger lever **11**A with a finger of the right hand holding the grip **3**B. This causes the rotation output unit **5** to rotate

in a rotation direction indicated by the arrow in FIG. **7**, with the drill bit **19** being attached.

(84) With the lower joint **23**D in the additional handle **23** in contact with the ground, the operator presses the front end **19**U of the rotating drill bit **19** against the drill target in front of the earth auger **1**. This allows drilling into the drill target to form a lateral hole in the drill target. The operator pushes the earth auger **1** forward with the drill bit **19** rotating. The lower joint **23**D may slide on the ground. This allows the earth auger **1** to move forward smoothly. To discharge a drilled material from the lateral hole formed in the drill target, the operator pulls out the earth auger **1** backward. The earth auger **1** moves backward smoothly as well.

(85) During a drilling operation into the drill target, a large reaction force may act on the gear housing **4** through the drill bit **19** and the rotation output unit **5**. When the drill bit **19** rotates in the rotation direction indicated by the arrow in FIG. **7**, a reaction force may act to cause the upper end of the body **6** to fall leftward.

(86) The additional handle **23** is at least partially located laterally from the gear housing **4** to receive a reaction force transmitted from the rotation output unit **5** to the gear housing **4**. In the present embodiment, the additional handle **23** is located leftward from the gear housing **4**. The grip portion **23**E in the additional handle **23** is gripped by the operator, and the lower joint **23**D in the additional handle **23** is in contact with the ground. The additional handle **23** thus receives a reaction force acting on the gear housing **4**.

Third Embodiment

(87) A third embodiment will now be described. The same or corresponding components as those in the above embodiments are given the same reference numerals herein and will be described briefly or will not be described. FIG. **8** is a perspective view of an earth auger **1** according to the present embodiment, describing its use. In the present embodiment, the contact member **22** and the additional handle **23**, which are described in the above embodiments, are fastened to the gear housing **4**.

(88) In the present embodiment, the left portion of the contact member **22** is located below the additional handle **23**. In the present embodiment, the contact member **22** protrudes leftward more largely from the left surface of the gear housing **4** than the contact member **22** in the first embodiment described above. The additional handle **23** is not in contact with the ground.

(89) In the present embodiment, the contact member **22** protrudes leftward largely from the left surface of the gear housing **4**. The contact member **22** thus receives a reaction force acting on the gear housing **4** with the left portion of the contact member **22** in contact with the ground. The operator grips the grip portion **23**E in the additional handle **23** located leftward from gear housing **4**. The additional handle **23** thus receives a reaction force acting on the gear housing **4**.

Fourth Embodiment

(90) A fourth embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.

(91) FIG. **9** is a front view of an earth auger **1** according to the present embodiment, describing its use. FIG. **10** is a perspective view of the earth auger **1** according to the present embodiment, describing its use. The earth auger **1** according to the present embodiment is used in a lateral position with the motor housing **2** located laterally (rightward) from the gear housing **4**. The body **6** receives the contact member **22** and the additional handle **23**, which are described in the above embodiments.

(92) The additional handle **23** is located above the gear housing **4** in the lateral position with the motor housing **2** located laterally from the gear housing **4**. The additional handle **23** is fastened to the gear housing **4**. The contact member **22** is fastened to the handle housing **3**.

(93) The operator grips the grip portion **23**E in the additional handle **23** with the right hand and the grip **3**B in the handle housing **3** with the left hand. The operator operates the trigger lever **11**A with a finger of the left hand holding the grip **3**B. This causes the rotation output unit **5** to rotate in a

rotation direction indicated by the arrow in FIG. **10**, with the drill bit **19** being attached.
(94) With the contact member **22** in contact with the ground, the operator presses the front end **19**U of the rotating drill bit **19** against the drill target in front of the earth auger **1**. This allows drilling into the drill target to form a lateral hole in the drill target. The operator pushes the earth auger **1** forward with the drill bit **19** rotating. The contact member **22** slides on the ground. This allows the earth auger **1** to move forward smoothly. To discharge a drilled material from the lateral hole formed in the drill target, the operator pulls out the earth auger **1** backward. The earth auger **1** moves backward smoothly as well.
(95) In the present embodiment, the additional handle **23** is located above the gear housing **4**. The operator grips the grip portion **23**E in the additional handle **23**. The additional handle **23** thus receives a reaction force acting on the gear housing **4**. The contact member **22** located leftward from the gear housing **4** is in contact with the ground. The contact member **22** thus receives a reaction force acting on the gear housing **4**.
Fifth Embodiment
(96) A fifth embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.
(97) FIG. **11** is a perspective view of an earth auger **1** according to the present embodiment, describing its use. FIG. **12** is a perspective view of an additional handle **24** in the present embodiment. In the present embodiment, the additional handle **24** is loop-shaped and is fastened to the left portion of the gear housing **4**. The additional handle **24** includes a facing portion **24**A, a joint **24**F, a straight portion **24**D, and a hand guard **24**E. The facing portion **24**A faces the left surface of the gear housing **4**. The joint **24**F is located leftward from the facing portion **24**A. The straight portion **24**D connects an upper portion of the facing portion **24**A to an upper portion of the joint **24**F. The hand guard **24**E connects a lower portion of the facing portion **24**A to a lower portion of the joint **24**F. The straight portion **24**D extends laterally (leftward) from the gear housing **4**. The hand guard **24**E is located below the straight portion **24**D. The joint **24**F has a lower end **24**G located below the hand guard **24**E.
(98) The facing portion **24**A has a screw opening **24**B and a screw opening **24**C. A screw placed in the screw opening **24**B is screwed into the threaded hole **16**. A screw placed in the screw opening **24**C is screwed into the threaded hole **18**. The additional handle **24** is thus fastened to the gear housing **4**.
(99) The earth auger **1** according to the present embodiment is used in a vertical position with the motor housing **2** located above the gear housing **4**. In this state, the lower end of the gear housing **4** is flush with at least a part of the additional handle **24**. In the embodiment, the lower end of the gear housing **4** is flush with the lower end **24**G of the joint **24**F. With the lower end of the gear housing **4** in contact with the ground, the additional handle **24** is at least partially in contact with the ground. In the embodiment, the lower end **24**G of the joint **24**F is in contact with the ground.
(100) The operator grips the straight portion **24**D in the additional handle **24** with the left hand and the grip **3**B in the handle housing **3** with the right hand. The operator operates the trigger lever **11**A with a finger of the right hand holding the grip **3**B. This causes the rotation output unit **5** to rotate in a rotation direction indicated by the arrow in FIG. **11**, with the drill bit **19** being attached.
(101) With the lower end **24**G of the joint **24**F in contact with the ground, the operator presses the front end **19**U of the rotating drill bit **19** against the drill target in front of the earth auger **1**. This allows drilling into the drill target to form a lateral hole in the drill target. The operator pushes the earth auger **1** forward with the drill bit **19** rotating. The lower end **24**G of the joint **24**F slides on the ground. This allows the earth auger **1** to move forward smoothly. To discharge a drilled material from the lateral hole formed in the drill target, the operator pulls out the earth auger **1** backward. The earth auger **1** moves backward smoothly as well.
(102) In the present embodiment, the additional handle **24** is located leftward from the gear housing

4. The straight portion **24**D in the additional handle **24** is gripped by the operator, and the lower end **24**G of the joint **24**F is in contact with the ground. The additional handle **24** thus receives a reaction force acting on the gear housing **4**. The hand guard **24**E prevents the operator's left hand from coming in contact with the ground. When the additional handle **24** receives a reaction force acting on the gear housing **4**, the hand guard **24**E prevents the operator's hand (left hand) from being caught between the ground and the straight portion **24**D.
(103) The lower end **24**G in the present embodiment protrudes downward from the left end of the hand guard **24**E. The lower end **24**G of the additional handle **24** is located farthest from the rotation output unit **5**. The lower end **24**G in contact with the ground fully receives a reaction force acting on the gear housing **4**. The lower end **24**G has a small contact area with the ground. The earth auger **1** thus moves smoothly in the front-rear direction.
Sixth Embodiment
(104) A sixth embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.
(105) FIG. **13** is a front perspective view of an earth auger **1** according to the present embodiment. In the present embodiment, a loop-shaped additional handle **25** is fastened to the left portion of the gear housing **4**. The additional handle **25** includes a facing portion **25**A, a bend **25**D, and a straight portion **25**E. The facing portion **25**A faces the left surface of the gear housing **4**. The bend **25**D is connected to the upper end of the facing portion **25**A. The straight portion **25**E is connected to a lower portion of the facing portion **25**A. The bend **25**D is curved downward at a farther distance leftward from the gear housing **4**. The straight portion **25**E extends laterally (leftward) from the gear housing **4** below the bend **25**D. The straight portion **25**E has its left end connected to a part of the bend **25**D. The bend **25**D has a lower end **25**F located below the straight portion **25**E. The operator can grip at least either a part of the bend **25**D or a part of the straight portion **25**E.
(106) The facing portion **25**A has a screw opening **25**B and a screw opening **25**C. A screw placed in the screw opening **25**B is screwed into the threaded hole **16**. A screw placed in the screw opening **25**C is screwed into the threaded hole **18**. The additional handle **25** is thus fastened to the gear housing **4**.
(107) The earth auger **1** according to the present embodiment is used in a vertical position with the motor housing **2** located above the gear housing **4**. In this state, the lower end of the gear housing **4** is flush with at least a part of the additional handle **25**. In the embodiment, the lower end of the gear housing **4** is flush with the lower end **25**F of the bend **25**D. With the lower end of the gear housing **4** in contact with the ground, the additional handle **25** is at least partially in contact with the ground. In the embodiment, the lower end **25**F of the bend **25**D is in contact with the ground.
(108) The operator grips the bend **25**D in the additional handle **25** with the left hand and the grip **3**B in the handle housing **3** with the right hand. The operator may grip the straight portion **25**E in the additional handle **25** with the left hand. The operator operates the trigger lever **11**A with a finger of the right hand holding the grip **3**B. This causes the rotation output unit **5** to rotate in a rotation direction indicated by the arrow in FIG. **13**, with the drill bit **19** being attached.
(109) With the lower end **25**F of the bend **25**D in contact with the ground, the operator presses the front end **19**U of the rotating drill bit **19** against the drill target in front of the earth auger **1**. This allows drilling into the drill target to form a lateral hole in the drill target. The operator pushes the earth auger **1** forward with the drill bit **19** rotating. The lower end **25**F of the bend **25**D slides on the ground. This allows the earth auger **1** to move forward smoothly. To discharge a drilled material from the lateral hole formed in the drill target, the operator pulls out the earth auger **1** backward. The earth auger **1** moves backward smoothly as well.
(110) In the present embodiment, the lower end **25**F has a small contact area with the ground. The earth auger **1** thus moves smoothly in the front-rear direction.
(111) In the present embodiment, the additional handle **25** is located leftward from the gear housing

4. The bend **25**D in the additional handle **25** is gripped by the operator, and the lower end **25**F of the bend **25**D in the additional handle **25** is in contact with the ground. The additional handle **25** thus receives a reaction force acting on the gear housing **4**.
REFERENCE SIGNS LIST
(112) **1** earth auger **2** motor housing **2**E outlet **3** handle housing **3**A front portion **3**B grip **3**C controller compartment **3**D battery connector **3**F inlet **4** gear housing **4**A cover **5** rotation output unit **5**A adapter **5**B fastener **6** body **7** battery mount **8** motor **9** forward-reverse switch lever **10** main switch **11** trigger switch **11**A trigger lever **11**B switch circuit **13** controller **14** reducer **15** speed switch lever **16** threaded hole **17** battery pack **18** threaded hole **19** drill bit **19**A drill shaft **19**B double-helix drill blade **19**C tip bit **19**D cutting blade **19**U front end **20** additional handle **20**A lateral rod **20**B vertical rod **20**C insertion portion **21** handle joint **21**A threaded hole **21**B screw opening **21**C screw opening **21**D screw opening **21**E screw opening **22** contact member **22**A contact surface **22**B front surface **22**C insertion portion **22**D threaded hole **23** additional handle **23**A screw opening **23**B facing portion **23**C upper joint **23**D lower joint **23**E grip portion **24** additional handle **24**A facing portion **24**B screw opening **24**C screw opening **24**D straight portion **24**E hand guard **24**F joint **24**G lower end **25** additional handle **25**A facing portion **25**B screw opening **25**C screw opening **25**D bend **25**E straight portion **25**F lower end **51** spindle **52** drill chuck **52**A insertion hole **53** needle bearing **54** ball bearing **55** bevel gear **81** stator **81**A stator core **81**B first insulator **81**C second insulator **81**D coil **81**E sensor circuit board **81**F connection wire **82** rotor **82**A rotor shaft **82**B rotor core **82**C permanent magnet **83** bearing **84** bearing **85** centrifugal fan **141** first planetary gear assembly **141**C first carrier **141**P planetary gear **141**R internal gear **141**S pinion gear **142** second planetary gear assembly **142**C second carrier **142**P planetary gear **142**R internal gear **142**S sun gear **143** countershaft **144** output shaft **145** support pin **146** support pin **147** bearing **148** bevel gear **150** switching member **150**H hole **151** connector **151**H hole **211** base **212** fixing portion

Claims

1. An earth auger, comprising: a motor; a motor housing accommodating the motor; a handle housing including a grip including a trigger switch operable to activate the motor; a reducer; a gear housing accommodating the reducer; a rotation output unit protruding frontward from the gear housing, the rotation output unit being rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit; and a handle fastened to the gear housing and at least partially located laterally from the gear housing, wherein with the motor housing located above the gear housing, the gear housing has a lower end flush with at least a part of the handle, and with the lower end of the gear housing in contact with a ground, the handle is at least partially in contact with the ground.
2. The earth auger according to claim 1, wherein the handle is at least partially located laterally from the gear housing to receive a reaction force transmitted from the rotation output unit to the gear housing.
3. The earth auger according to claim 1, wherein the handle is loop-shaped.
4. The earth auger according to claim 1, wherein the handle includes a straight portion extending laterally from the gear housing, and a hand guard located below the straight portion.
5. The earth auger according to claim 1, wherein the handle includes a bend curved downward at a farther distance from the gear housing, and a straight portion extending laterally from the gear housing below the bend.
6. The earth auger according to claim 1, further comprising: a contact member located below of the rotation output unit.
7. The earth auger according to claim 6, wherein the contact member is at least partially located laterally from the gear housing.

8. The earth auger according to claim 6, wherein the contact member is fastened to the gear housing.

9. An earth auger, comprising: a motor; a motor housing accommodating the motor; a handle housing including a grip including a trigger switch operable to activate the motor; a reducer; a gear housing accommodating the reducer; a rotation output unit protruding frontward from the gear housing, the rotation output unit being rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit; and a handle fastened to the gear housing and at least partially located laterally from the gear housing, wherein with a lower end of the gear housing in contact with a ground, the handle is at least partially in contact with the ground, and the handle includes a straight portion extending laterally from the gear housing, and a hand guard located below the straight portion.

10. An earth auger, comprising: a motor; a motor housing accommodating the motor; a handle housing including a grip including a trigger switch operable to activate the motor, the handle housing disposed above the motor housing; a reducer; a gear housing accommodating the reducer, the gear housing disposed below the motor housing; a rotation output unit protruding frontward from the gear housing, the rotation output unit being rotatable under a rotational force transmitted from the motor through the reducer with a drill bit being attached to the rotation output unit; and a loop-shaped handle disposed on a right side or a left side of the gear housing, the loop-shaped handle including an additional grip extending in a vertical direction.